

ArogyaSetu Bridge — Database Architecture

25 POSTGRESQL
TABLES

ArogyaSetu Bridge • SIH 2026 Problem Statement ID 26133

ArogyaSetu Bridge — Database Schema & ER Diagrams

SIH 2026 | Problem Statement 26133

1. Database Design Philosophy

The database design for ArogyaSetu Bridge follows these principles:

- **FHIR R4 Alignment:** Core clinical tables mirror HL7 FHIR R4 resource structures (Patient, Encounter, Observation, MedicationRequest, ServiceRequest) to ensure seamless ABDM interoperability.
- **Relational Integrity:** PostgreSQL with strict foreign key relationships for patient safety — no orphaned records, no dangling references.
- **JSONB Flexibility:** Clinical observations and triage responses use JSONB columns for semi-structured data that varies by condition type, while maintaining indexable structured columns for all query-critical fields.
- **Audit Trail:** Every table includes `created_at`, `updated_at`, `created_by`, and `updated_by` columns. Soft-delete pattern (`deleted_at`) prevents accidental data loss.
- **Multi-Tenancy:** Facility-scoped data with `facility_id` foreign keys on all clinical tables, enabling both facility-level isolation and cross-facility queries for referral tracking.

2. Complete Database Schema

2.1 Core Identity & Access Tables

Table: `users`

Stores all system users — health workers, doctors, specialists, pharmacists, administrators.

Column	Type	Constraints	Description
id	UUID	PK, DEFAULT gen_random_uuid()	Unique user identifier
abha_id	VARCHAR(14)	UNIQUE, NULLABLE	ABHA health ID (for patients who are also users)
full_name	VARCHAR(255)	NOT NULL	Full name of the user
phone	VARCHAR(15)	NOT NULL, UNIQUE	Mobile number (primary auth)
email	VARCHAR(255)	NULLABLE	Email address
role	ENUM	NOT NULL	One of: ASHA, ANM, CHO, DOCTOR, SPECIALIST, PHARMACIST, DHO, ADMIN
facility_id	UUID	FK → facilities.id, NOT NULL	Primary assigned facility
specialization	VARCHAR(100)	NULLABLE	Medical specialization (for DOCTOR/SPECIALIST roles)
license_number	VARCHAR(50)	NULLABLE	Medical license / registration number
language_pref	VARCHAR(10)	DEFAULT 'mr'	Preferred language (mr/hi/en)
is_active	BOOLEAN	DEFAULT TRUE	Account active status
last_login_at	TIMESTAMPTZ	NULLABLE	Last successful login timestamp
created_at	TIMESTAMPTZ	DEFAULT NOW()	Record creation time
updated_at	TIMESTAMPTZ	DEFAULT NOW()	Last update time
deleted_at	TIMESTAMPTZ	NULLABLE	Soft delete timestamp

Indexes: idx_users_phone , idx_users_facility , idx_users_role , idx_users_abha

Table: facilities

Registry of all healthcare facilities in the network — Sub-Centres, PHCs, CHCs, District Hospitals.

Column	Type	Constraints	Description
id	UUID	PK	Unique facility identifier
hfr_id	VARCHAR(50)	UNIQUE, NULLABLE	Health Facility Registry ID (ABDM)
name	VARCHAR(255)	NOT NULL	Facility name
type	ENUM	NOT NULL	SUB_CENTRE, PHC, CHC, SUB_DISTRICT_HOSPITAL, DISTRICT_HOSPITAL, MEDICAL_COLLEGE
address	TEXT	NOT NULL	Full address
district	VARCHAR(100)	NOT NULL	District name
taluka	VARCHAR(100)	NOT NULL	Taluka / sub-district name
state	VARCHAR(50)	DEFAULT 'Maharashtra'	State
pin_code	VARCHAR(6)	NOT NULL	PIN code
latitude	DECIMAL(10,8)	NULLABLE	GPS latitude
longitude	DECIMAL(11,8)	NULLABLE	GPS longitude
parent_facility_id	UUID	FK → facilities.id, NULLABLE	Referral parent (e.g., PHC's parent is CHC)
phone	VARCHAR(15)	NULLABLE	Facility contact number
operating_hours	JSONB	NULLABLE	Operating schedule by day
services_available	TEXT[]	NULLABLE	Array of available services
bed_count	INT	DEFAULT 0	Number of beds
has_teleconsult	BOOLEAN	DEFAULT FALSE	Teleconsultation capability
is_active	BOOLEAN	DEFAULT TRUE	Facility active status
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

Indexes: idx_facilities_type , idx_facilities_district , idx_facilities_parent , idx_facilities_location (GIST on lat/lng)

Table: user_sessions

JWT session management and audit.

Column	Type	Constraints	Description
id	UUID	PK	Session identifier
user_id	UUID	FK → users.id, NOT NULL	
token_hash	VARCHAR(64)	NOT NULL	SHA-256 hash of JWT
device_info	JSONB	NULLABLE	Device fingerprint data
ip_address	INET	NULLABLE	
expires_at	TIMESTAMPTZ	NOT NULL	Token expiry
created_at	TIMESTAMPTZ	DEFAULT NOW()	
revoked_at	TIMESTAMPTZ	NULLABLE	Manual revocation timestamp

2.2 Patient Management Tables

Table: `patients`

Core patient demographic and identity record.

Column	Type	Constraints	Description
id	UUID	PK	Internal patient identifier
abha_id	VARCHAR(14)	UNIQUE, NULLABLE	ABHA health ID
abha_address	VARCHAR(255)	NULLABLE	ABHA address (name@abdm)
full_name	VARCHAR(255)	NOT NULL	Patient full name
date_of_birth	DATE	NOT NULL	Date of birth
gender	ENUM	NOT NULL	MALE, FEMALE, OTHER
phone	VARCHAR(15)	NULLABLE	Patient or guardian phone
guardian_name	VARCHAR(255)	NULLABLE	Guardian name (for minors)
guardian_phone	VARCHAR(15)	NULLABLE	Guardian contact
address_line	TEXT	NULLABLE	Address
village	VARCHAR(100)	NULLABLE	Village name
taluka	VARCHAR(100)	NULLABLE	Taluka
district	VARCHAR(100)	NULLABLE	District
pin_code	VARCHAR(6)	NULLABLE	PIN code
blood_group	VARCHAR(5)	NULLABLE	Blood group
allergies	TEXT[]	NULLABLE	Known allergies
chronic_conditions	TEXT[]	NULLABLE	Known chronic conditions
is_high_risk	BOOLEAN	DEFAULT FALSE	High-risk flag
high_risk_reason	TEXT	NULLABLE	Reason for high-risk classification
preferred_language	VARCHAR(10)	DEFAULT 'mr'	mr/hi/en
registered_facility_id	UUID	FK → facilities.id	Facility where patient was first registered
registered_by	UUID	FK → users.id	User who registered the patient
consent_status	ENUM	DEFAULT 'PENDING'	PENDING, GRANTED, REVOKED
consent_granted_at	TIMESTAMPTZ	NULLABLE	When consent was granted
consent_expiry	TIMESTAMPTZ	NULLABLE	Consent expiry date
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	
deleted_at	TIMESTAMPTZ	NULLABLE	

Indexes: `idx_patients_abha`, `idx_patients_phone`, `idx_patients_name` (trigram for fuzzy search), `idx_patients_village_district`, `idx_patients_high_risk`

Table: `patient_vitals`

Records of vital sign measurements taken at any facility or during home visits.

Column	Type	Constraints	Description
<code>id</code>	UUID	PK	
<code>patient_id</code>	UUID	FK → patients.id, NOT NULL	
<code>recorded_by</code>	UUID	FK → users.id, NOT NULL	Health worker who recorded
<code>facility_id</code>	UUID	FK → facilities.id, NOT NULL	Where vitals were recorded
<code>recorded_at</code>	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Measurement time
<code>blood_pressure_systolic</code>	INT	NULLABLE	mmHg
<code>blood_pressure_diastolic</code>	INT	NULLABLE	mmHg
<code>heart_rate</code>	INT	NULLABLE	bpm
<code>temperature</code>	DECIMAL(4,1)	NULLABLE	Celsius
<code>spo2</code>	INT	NULLABLE	Oxygen saturation %
<code>respiratory_rate</code>	INT	NULLABLE	breaths/min
<code>weight</code>	DECIMAL(5,2)	NULLABLE	kg
<code>height</code>	DECIMAL(5,2)	NULLABLE	cm
<code>bmi</code>	DECIMAL(4,1)	NULLABLE	Calculated BMI
<code>blood_sugar_fasting</code>	DECIMAL(5,1)	NULLABLE	mg/dL
<code>blood_sugar_pp</code>	DECIMAL(5,1)	NULLABLE	mg/dL (post-prandial)
<code>notes</code>	TEXT	NULLABLE	Additional observations
<code>created_at</code>	TIMESTAMPTZ	DEFAULT NOW()	

Indexes: `idx_vitals_patient_date`, `idx_vitals_facility`

2.3 Clinical & Encounter Tables

Table: `encounters`

Every clinical interaction — consultation, teleconsultation, home visit, emergency.

Column	Type	Constraints	Description
id	UUID	PK	Encounter identifier
patient_id	UUID	FK → patients.id, NOT NULL	
facility_id	UUID	FK → facilities.id, NOT NULL	Facility of encounter
provider_id	UUID	FK → users.id, NOT NULL	Attending doctor/CHO
encounter_type	ENUM	NOT NULL	OPD, TELECONSULT_VIDEO, TELECONSULT_ASYNC, HOME_VISIT, EMERGENCY, FOLLOW_UP
status	ENUM	NOT NULL, DEFAULT 'SCHEDULED'	SCHEDULED, IN_PROGRESS, COMPLETED, CANCELLED, NO_SHOW
scheduled_at	TIMESTAMPTZ	NULLABLE	Scheduled time
started_at	TIMESTAMPTZ	NULLABLE	Actual start
ended_at	TIMESTAMPTZ	NULLABLE	Actual end
chief_complaint	TEXT	NULLABLE	Primary complaint
clinical_notes	TEXT	NULLABLE	Doctor's clinical notes
diagnosis_icd_codes	JSONB	NULLABLE	Array of ICD-10/11 codes with descriptions
diagnosis_snomed_codes	JSONB	NULLABLE	Array of SNOMED CT codes
follow_up_required	BOOLEAN	DEFAULT FALSE	
follow_up_date	DATE	NULLABLE	Scheduled follow-up date
triage_id	UUID	FK → triage_assessments.id, NULLABLE	Linked triage assessment
referral_id	UUID	FK → referrals.id, NULLABLE	If this encounter resulted from a referral
teleconsult_session_id	VARCHAR(100)	NULLABLE	WebRTC session ID
fhir_resource_id	VARCHAR(100)	NULLABLE	FHIR Encounter resource ID
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

Indexes: idx_encounters_patient, idx_encounters_provider, idx_encounters_facility_date, idx_encounters_status, idx_encounters_type

Table: `triage_assessments`

Digital triage records — symptom assessment and urgency classification.

Column	Type	Constraints	Description
<code>id</code>	UUID	PK	
<code>patient_id</code>	UUID	FK → patients.id, NOT NULL	
<code>assessed_by</code>	UUID	FK → users.id, NOT NULL	ASHA/ANM/CHO who conducted triage
<code>facility_id</code>	UUID	FK → facilities.id, NOT NULL	
<code>assessed_at</code>	TIMESTAMPTZ	DEFAULT NOW()	
<code>symptoms</code>	JSONB	NOT NULL	Structured symptom checklist responses
<code>symptom_duration_days</code>	INT	NULLABLE	How long symptoms have been present
<code>urgency_level</code>	ENUM	NOT NULL	EMERGENCY, URGENT, SEMI_URGENT, NON_URGENT
<code>recommended_action</code>	ENUM	NOT NULL	EMERGENCY_TRANSFER, REFER_TO_PHC, REFER_TO_CHC, TELECONSULT, SELF_CARE
<code>recommended_facility_id</code>	UUID	FK → facilities.id, NULLABLE	System-recommended facility
<code>triage_algorithm_version</code>	VARCHAR(20)	NOT NULL	Algorithm version used
<code>confidence_score</code>	DECIMAL(3,2)	NULLABLE	0.00 to 1.00
<code>override_by</code>	UUID	FK → users.id, NULLABLE	If a doctor overrode the triage
<code>override_reason</code>	TEXT	NULLABLE	
<code>created_at</code>	TIMESTAMPTZ	DEFAULT NOW()	

Indexes: `idx_triage_patient` , `idx_triage_urgency` , `idx_triage_facility_date`

Table: `prescriptions`

Medication prescriptions linked to encounters.

Column	Type	Constraints	Description
id	UUID	PK	
encounter_id	UUID	FK → encounters.id, NOT NULL	
patient_id	UUID	FK → patients.id, NOT NULL	
prescribed_by	UUID	FK → users.id, NOT NULL	
facility_id	UUID	FK → facilities.id, NOT NULL	
prescribed_at	TIMESTAMPTZ	DEFAULT NOW()	
is_active	BOOLEAN	DEFAULT TRUE	
notes	TEXT	NULLABLE	General prescription notes
fhir_resource_id	VARCHAR(100)	NULLABLE	
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

Table: prescription_items

Individual medication items within a prescription.

Column	Type	Constraints	Description
id	UUID	PK	
prescription_id	UUID	FK → prescriptions.id, NOT NULL	
medicine_id	UUID	FK → medicines.id, NULLABLE	Link to medicine catalog
medicine_name	VARCHAR(255)	NOT NULL	Medicine name (free text or from catalog)
dosage	VARCHAR(100)	NOT NULL	e.g., "500mg"
frequency	VARCHAR(100)	NOT NULL	e.g., "1-0-1" (morning-afternoon-night)
duration_days	INT	NOT NULL	Duration in days
route	VARCHAR(50)	DEFAULT 'ORAL'	ORAL, IV, IM, TOPICAL, etc.
instructions	TEXT	NULLABLE	Special instructions
is_dispensed	BOOLEAN	DEFAULT FALSE	Whether dispensed by pharmacy
dispensed_at	TIMESTAMPTZ	NULLABLE	
dispensed_by	UUID	FK → users.id, NULLABLE	

2.4 Referral Management Tables

Table: referrals

End-to-end referral tracking from initiation to completion.

Column	Type	Constraints	Description
id	UUID	PK	Referral identifier
referral_code	VARCHAR(20)	UNIQUE, NOT NULL	Human-readable referral code (e.g., REF-MH-2026-0001)
patient_id	UUID	FK → patients.id, NOT NULL	
source_facility_id	UUID	FK → facilities.id, NOT NULL	Referring facility
source_provider_id	UUID	FK → users.id, NOT NULL	Referring doctor/CHO
target_facility_id	UUID	FK → facilities.id, NOT NULL	Receiving facility
target_provider_id	UUID	FK → users.id, NULLABLE	Assigned specialist (if specific)
encounter_id	UUID	FK → encounters.id, NULLABLE	Source encounter
triage_id	UUID	FK → triage_assessments.id, NULLABLE	Linked triage
referral_reason	TEXT	NOT NULL	Clinical reason for referral
urgency	ENUM	NOT NULL	EMERGENCY, URGENT, ROUTINE
status	ENUM	NOT NULL, DEFAULT 'INITIATED'	INITIATED, ACCEPTED, SCHEDULED, IN_TRANSIT, ARRIVED, CONSULTATION_DONE, COMPLETED, REJECTED, ABANDONED
status_history	JSONB	DEFAULT '[]'	Array of
clinical_summary	TEXT	NULLABLE	Summary for receiving provider
attachments	JSONB	NULLABLE	Array of attached report/image URLs
scheduled_date	DATE	NULLABLE	Scheduled appointment at target facility
completed_at	TIMESTAMPTZ	NULLABLE	When referral was completed
patient_feedback	TEXT	NULLABLE	Patient feedback on referral experience
patient_feedback_rating	INT	NULLABLE	1-5 rating
reminder_count	INT	DEFAULT 0	Number of reminders sent
last_reminder_at	TIMESTAMPTZ	NULLABLE	
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

Indexes: `idx_referrals_patient`, `idx_referrals_source`, `idx_referrals_target`, `idx_referrals_status`, `idx_referrals_code`, `idx_referrals_urgency`

2.5 Diagnostic & Lab Tables

Table: `diagnostic_orders`

Lab test and imaging orders.

Column	Type	Constraints	Description
<code>id</code>	UUID	PK	
<code>encounter_id</code>	UUID	FK → encounters.id, NOT NULL	
<code>patient_id</code>	UUID	FK → patients.id, NOT NULL	
<code>ordered_by</code>	UUID	FK → users.id, NOT NULL	
<code>ordering_facility_id</code>	UUID	FK → facilities.id, NOT NULL	Where the order was placed
<code>performing_facility_id</code>	UUID	FK → facilities.id, NULLABLE	Where the test will be done
<code>test_name</code>	VARCHAR(255)	NOT NULL	
<code>test_code_loinc</code>	VARCHAR(20)	NULLABLE	LOINC code
<code>test_category</code>	ENUM	NOT NULL	LAB, RADIOLOGY, PATHOLOGY, OTHER
<code>priority</code>	ENUM	DEFAULT 'ROUTINE'	STAT, URGENT, ROUTINE
<code>status</code>	ENUM	DEFAULT 'ORDERED'	ORDERED, SAMPLE_COLLECTED, IN_PROGRESS, COMPLETED, CANCELLED
<code>ordered_at</code>	TIMESTAMPTZ	DEFAULT NOW()	
<code>completed_at</code>	TIMESTAMPTZ	NULLABLE	
<code>notes</code>	TEXT	NULLABLE	
<code>created_at</code>	TIMESTAMPTZ	DEFAULT NOW()	
<code>updated_at</code>	TIMESTAMPTZ	DEFAULT NOW()	

Table: `diagnostic_results`

Results of completed diagnostic orders.

Column	Type	Constraints	Description
id	UUID	PK	
order_id	UUID	FK → diagnostic_orders.id, NOT NULL	
patient_id	UUID	FK → patients.id, NOT NULL	
reported_by	UUID	FK → users.id, NULLABLE	Lab technician / radiologist
facility_id	UUID	FK → facilities.id, NOT NULL	
result_data	JSONB	NOT NULL	Structured result values
result_text	TEXT	NULLABLE	Free-text interpretation
is_abnormal	BOOLEAN	DEFAULT FALSE	Flagged as abnormal
abnormal_flags	JSONB	NULLABLE	Specific abnormal values
report_file_url	VARCHAR(500)	NULLABLE	URL to report PDF/image
fhir_resource_id	VARCHAR(100)	NULLABLE	
reported_at	TIMESTAMPTZ	DEFAULT NOW()	
created_at	TIMESTAMPTZ	DEFAULT NOW()	

Table: facility_equipment

Registry of diagnostic equipment available at each facility.

Column	Type	Constraints	Description
id	UUID	PK	
facility_id	UUID	FK → facilities.id, NOT NULL	
equipment_name	VARCHAR(255)	NOT NULL	
equipment_type	ENUM	NOT NULL	XRAY, ULTRASOUND, ECG, BLOOD_ANALYZER, URINE_ANALYZER, OTHER
manufacturer	VARCHAR(255)	NULLABLE	
is_operational	BOOLEAN	DEFAULT TRUE	Current status
last_maintenance_date	DATE	NULLABLE	
tests_supported	TEXT[]	NULLABLE	List of tests this equipment can perform
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

2.6 Medicine & Pharmacy Tables

Table: medicines

Master medicine catalog.

Column	Type	Constraints	Description
id	UUID	PK	
generic_name	VARCHAR(255)	NOT NULL	Generic / salt name
brand_name	VARCHAR(255)	NULLABLE	Common brand name
form	ENUM	NOT NULL	TABLET, CAPSULE, SYRUP, INJECTION, CREAM, DROPS, OTHER
strength	VARCHAR(50)	NOT NULL	e.g., "500mg", "250mg/5ml"
category	VARCHAR(100)	NULLABLE	Therapeutic category
is_essential	BOOLEAN	DEFAULT FALSE	On Essential Drug List
requires_prescription	BOOLEAN	DEFAULT TRUE	
created_at	TIMESTAMPTZ	DEFAULT NOW()	

Table: `medicine_stock`

Real-time stock levels at each facility.

Column	Type	Constraints	Description
<code>id</code>	UUID	PK	
<code>facility_id</code>	UUID	FK → facilities.id, NOT NULL	
<code>medicine_id</code>	UUID	FK → medicines.id, NOT NULL	
<code>current_quantity</code>	INT	NOT NULL, DEFAULT 0	Current stock count
<code>unit</code>	VARCHAR(20)	NOT NULL	TABLETS, BOTTLES, VIALS, STRIPS, etc.
<code>reorder_level</code>	INT	DEFAULT 50	Alert threshold
<code>batch_number</code>	VARCHAR(50)	NULLABLE	
<code>expiry_date</code>	DATE	NULLABLE	
<code>last_restocked_at</code>	TIMESTAMPTZ	NULLABLE	
<code>last_updated_by</code>	UUID	FK → users.id	
<code>updated_at</code>	TIMESTAMPTZ	DEFAULT NOW()	

Indexes: `idx_stock_facility_medicine` (composite unique), `idx_stock_low` (WHERE `current_quantity` <= `reorder_level`), `idx_stock_expiry`

Table: `medicine_stock_transactions`

Audit trail of all stock movements (additions and dispensations).

Column	Type	Constraints	Description
<code>id</code>	UUID	PK	
<code>stock_id</code>	UUID	FK → medicine_stock.id, NOT NULL	
<code>transaction_type</code>	ENUM	NOT NULL	RECEIVED, DISPENSED, EXPIRED, DAMAGED, TRANSFERRED, ADJUSTED
<code>quantity</code>	INT	NOT NULL	Positive for additions, negative for reductions
<code>reference_id</code>	UUID	NULLABLE	Link to prescription_item or transfer record
<code>performed_by</code>	UUID	FK → users.id, NOT NULL	
<code>notes</code>	TEXT	NULLABLE	
<code>created_at</code>	TIMESTAMPTZ	DEFAULT NOW()	

2.7 Queue & Appointment Tables

Table: appointments

Scheduled appointments at facilities.

Column	Type	Constraints	Description
id	UUID	PK	
patient_id	UUID	FK → patients.id, NOT NULL	
facility_id	UUID	FK → facilities.id, NOT NULL	
provider_id	UUID	FK → users.id, NULLABLE	Specific provider if assigned
appointment_type	ENUM	NOT NULL	OPD, TELECONSULT, FOLLOW_UP, VACCINATION, ANC_CHECKUP
scheduled_date	DATE	NOT NULL	
scheduled_time	TIME	NULLABLE	
token_number	INT	NULLABLE	Queue token
estimated_wait_minutes	INT	NULLABLE	
status	ENUM	DEFAULT 'BOOKED'	BOOKED, CHECKED_IN, IN_PROGRESS, COMPLETED, CANCELLED, NO_SHOW
booked_via	ENUM	DEFAULT 'APP'	APP, PHONE, WALK_IN, REFERRAL
reminder_sent	BOOLEAN	DEFAULT FALSE	
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

Indexes: idx_appointments_facility_date , idx_appointments_patient , idx_appointments_status

Table: queue_entries

Real-time queue management for walk-in and checked-in patients.

Column	Type	Constraints	Description
id	UUID	PK	
facility_id	UUID	FK → facilities.id, NOT NULL	
patient_id	UUID	FK → patients.id, NOT NULL	
appointment_id	UUID	FK → appointments.id, NULLABLE	
token_number	INT	NOT NULL	
counter	VARCHAR(20)	NULLABLE	Counter/room assignment
status	ENUM	DEFAULT 'WAITING'	WAITING, CALLED, IN_CONSULTATION, DONE, SKIPPED
priority	INT	DEFAULT 5	1 = highest priority (emergency)
entered_at	TIMESTAMPTZ	DEFAULT NOW()	
called_at	TIMESTAMPTZ	NULLABLE	
completed_at	TIMESTAMPTZ	NULLABLE	

2.8 Follow-Up & Notifications Tables

Table: follow_up_schedules

Automated follow-up tracking for high-risk patients and chronic conditions.

Column	Type	Constraints	Description
id	UUID	PK	
patient_id	UUID	FK → patients.id, NOT NULL	
encounter_id	UUID	FK → encounters.id, NULLABLE	
assigned_to	UUID	FK → users.id, NOT NULL	ASHA/ANM responsible
follow_up_type	ENUM	NOT NULL	ANC, PNC, CHRONIC_DISEASE, POST_SURGERY, IMMUNIZATION, GENERAL
scheduled_date	DATE	NOT NULL	
status	ENUM	DEFAULT 'PENDING'	PENDING, COMPLETED, OVERDUE, CANCELLED
priority	ENUM	DEFAULT 'NORMAL'	HIGH, NORMAL, LOW
notes	TEXT	NULLABLE	
completed_at	TIMESTAMPTZ	NULLABLE	
completed_notes	TEXT	NULLABLE	
reminder_count	INT	DEFAULT 0	
created_at	TIMESTAMPTZ	DEFAULT NOW()	
updated_at	TIMESTAMPTZ	DEFAULT NOW()	

Indexes: idx_followup_patient , idx_followup_assigned , idx_followup_status_date

Table: notifications

System-generated notifications for all user types.

Column	Type	Constraints	Description
id	UUID	PK	
recipient_user_id	UUID	FK → users.id, NULLABLE	For provider notifications
recipient_patient_id	UUID	FK → patients.id, NULLABLE	For patient notifications
recipient_phone	VARCHAR(15)	NULLABLE	Phone for SMS delivery
channel	ENUM	NOT NULL	SMS, WHATSAPP, PUSH, IN_APP
category	ENUM	NOT NULL	APPOINTMENT_REMINDER, REFERRAL_UPDATE, MEDICINE_REMINDER, STOCK_ALERT, FOLLOW_UP_DUE, EMERGENCY, SYSTEM
title	VARCHAR(255)	NOT NULL	
body	TEXT	NOT NULL	
language	VARCHAR(10)	DEFAULT 'mr'	
status	ENUM	DEFAULT 'QUEUED'	QUEUED, SENT, DELIVERED, FAILED, READ
sent_at	TIMESTAMPTZ	NULLABLE	
read_at	TIMESTAMPTZ	NULLABLE	
reference_type	VARCHAR(50)	NULLABLE	e.g., 'referral', 'appointment'
reference_id	UUID	NULLABLE	ID of referenced entity
created_at	TIMESTAMPTZ	DEFAULT NOW()	

2.9 Analytics & Audit Tables

Table: facility_daily_stats

Aggregated daily statistics for facility dashboards.

Column	Type	Constraints	Description
id	UUID	PK	
facility_id	UUID	FK → facilities.id, NOT NULL	
stat_date	DATE	NOT NULL	
total_patients_seen	INT	DEFAULT 0	
total_teleconsults	INT	DEFAULT 0	
total_referrals_sent	INT	DEFAULT 0	
total_referrals_received	INT	DEFAULT 0	
referrals_completed	INT	DEFAULT 0	
avg_wait_time_minutes	DECIMAL(5,1)	NULLABLE	
medicines_out_of_stock	INT	DEFAULT 0	
high_risk_patients_active	INT	DEFAULT 0	
follow_ups_completed	INT	DEFAULT 0	
follow_ups_overdue	INT	DEFAULT 0	
created_at	TIMESTAMPTZ	DEFAULT NOW()	

Indexes: idx_stats_facility_date (composite unique)

Table: audit_log

Immutable audit trail for all data access and modifications.

Column	Type	Constraints	Description
id	BIGSERIAL	PK	Auto-incrementing for ordering
user_id	UUID	FK → users.id, NULLABLE	
action	ENUM	NOT NULL	CREATE, READ, UPDATE, DELETE, LOGIN, LOGOUT, EXPORT, CONSENT_GRANT, CONSENT_REVOKE
resource_type	VARCHAR(50)	NOT NULL	Table/entity name
resource_id	UUID	NULLABLE	ID of affected record
facility_id	UUID	FK → facilities.id, NULLABLE	
ip_address	INET	NULLABLE	
details	JSONB	NULLABLE	Additional context
created_at	TIMESTAMPTZ	DEFAULT NOW()	

Indexes: idx_audit_user , idx_audit_resource , idx_audit_timestamp

2.10 Sync & Offline Tables

Table: sync_queue

Tracks pending data synchronizations from offline devices.

Column	Type	Constraints	Description
id	UUID	PK	
device_id	VARCHAR(100)	NOT NULL	Device identifier
user_id	UUID	FK → users.id, NOT NULL	
entity_type	VARCHAR(50)	NOT NULL	Table name
entity_id	UUID	NOT NULL	Record ID
operation	ENUM	NOT NULL	INSERT, UPDATE, DELETE
payload	JSONB	NOT NULL	Full record data
client_timestamp	TIMESTAMPTZ	NOT NULL	When change was made on device
server_timestamp	TIMESTAMPTZ	NULLABLE	When synced to server
sync_status	ENUM	DEFAULT 'PENDING'	PENDING, SYNCED, CONFLICT, FAILED
conflict_resolution	JSONB	NULLABLE	Resolution details if conflict
retry_count	INT	DEFAULT 0	
created_at	TIMESTAMPTZ	DEFAULT NOW()	

Indexes: idx_sync_device_status , idx_sync_entity

3. Entity-Relationship Diagrams

3.1 Main ER Diagram — Full System Overview

```
erDiagram
    FACILITIES ||--o{ USERS : "employs"
    FACILITIES ||--o{ PATIENTS : "registers"
    FACILITIES ||--o{ ENCOUNTERS : "hosts"
    FACILITIES ||--o{ MEDICINE_STOCK : "stores"
    FACILITIES ||--o{ FACILITY_EQUIPMENT : "has"
    FACILITIES ||--o{ APPOINTMENTS : "schedules"
    FACILITIES ||--o{ QUEUE_ENTRIES : "manages"
    FACILITIES ||--o{ FACILITY_DAILY_STATS : "reports"
    FACILITIES ||--o| FACILITIES : "parent_facility"

    USERS ||--o{ ENCOUNTERS : "provides_care"
    USERS ||--o{ TRIAGE_ASSESSMENTS : "assesses"
    USERS ||--o{ PRESCRIPTIONS : "prescribes"
    USERS ||--o{ REFERRALS : "initiates"
    USERS ||--o{ DIAGNOSTIC_ORDERS : "orders"
    USERS ||--o{ FOLLOW_UP_SCHEDULES : "assigned_to"
    USERS ||--o{ PATIENT_VITALS : "records"
    USERS ||--o{ USER_SESSIONS : "authenticates"

    PATIENTS ||--o{ ENCOUNTERS : "receives_care"
    PATIENTS ||--o{ PATIENT_VITALS : "has_vitals"
    PATIENTS ||--o{ TRIAGE_ASSESSMENTS : "triaged"
    PATIENTS ||--o{ PRESCRIPTIONS : "prescribed"
    PATIENTS ||--o{ REFERRALS : "referred"
    PATIENTS ||--o{ DIAGNOSTIC_ORDERS : "tested"
    PATIENTS ||--o{ APPOINTMENTS : "books"
    PATIENTS ||--o{ FOLLOW_UP_SCHEDULES : "tracked"
    PATIENTS ||--o{ NOTIFICATIONS : "notified"

    ENCOUNTERS ||--o{ PRESCRIPTIONS : "generates"
    ENCOUNTERS ||--o{ DIAGNOSTIC_ORDERS : "orders"
    ENCOUNTERS ||--o| TRIAGE_ASSESSMENTS : "linked_triage"
    ENCOUNTERS ||--o| REFERRALS : "from_referral"
    ENCOUNTERS ||--o{ FOLLOW_UP_SCHEDULES : "creates"

    PRESCRIPTIONS ||--o{ PRESCRIPTION_ITEMS : "contains"
    PRESCRIPTION_ITEMS ||--o| MEDICINES : "references"

    DIAGNOSTIC_ORDERS ||--o{ DIAGNOSTIC_RESULTS : "produces"

    MEDICINES ||--o{ MEDICINE_STOCK : "tracked_at"
    MEDICINE_STOCK ||--o{ MEDICINE_STOCK_TRANSACTIONS : "logs"

    REFERRALS }|--|| FACILITIES : "from_facility"
    REFERRALS }|--|| FACILITIES : "to_facility"

    APPOINTMENTS ||--o| QUEUE_ENTRIES : "generates"
```

3.2 Patient Care Journey ER Diagram

erDiagram

```
PATIENT ||--|| REGISTRATION : "registers_via"
REGISTRATION ||--|| ABHA_ID : "linked_to"

PATIENT ||--||{ HOME_VISIT : "receives"
HOME_VISIT ||--|| ASHA_WORKER : "conducted_by"
HOME_VISIT ||--|| VITALS_RECORD : "captures"
HOME_VISIT ||--o| SYMPTOM_SCREEN : "performs"

SYMPTOM_SCREEN ||--|| TRIAGE : "feeds_into"
TRIAGE ||--|| URGENCY_CLASSIFICATION : "produces"

URGENCY_CLASSIFICATION ||--o| SELF_CARE_ADVICE : "if_non_urgent"
URGENCY_CLASSIFICATION ||--o| TELECONSULTATION : "if_semi_urgent"
URGENCY_CLASSIFICATION ||--o| PHC_REFERRAL : "if_urgent"
URGENCY_CLASSIFICATION ||--o| EMERGENCY_TRANSFER : "if_emergency"

TELECONSULTATION ||--|| DOCTOR_AT_HUB : "connects_to"
TELECONSULTATION ||--|| PRESCRIPTION : "generates"
TELECONSULTATION ||--o| SPECIALIST_REFERRAL : "may_generate"

PHC_REFERRAL ||--|| APPOINTMENT : "creates"
APPOINTMENT ||--|| QUEUE_TOKEN : "assigned"
APPOINTMENT ||--|| ENCOUNTER : "becomes"
ENCOUNTER ||--|| PRESCRIPTION : "generates"
ENCOUNTER ||--o| DIAGNOSTIC_ORDER : "may_order"
ENCOUNTER ||--o| REFERRAL_TO_CHC : "may_refer"

DIAGNOSTIC_ORDER ||--|| DIAGNOSTIC_RESULT : "produces"
DIAGNOSTIC_RESULT ||--|| PATIENT_RECORD : "stored_in"

REFERRAL_TO_CHC ||--|| SPECIALIST_CONSULT : "leads_to"
SPECIALIST_CONSULT ||--|| TREATMENT_PLAN : "produces"
TREATMENT_PLAN ||--|| FOLLOW_UP_SCHEDULE : "creates"

FOLLOW_UP_SCHEDULE ||--|| ASHA_WORKER : "assigned_to"
FOLLOW_UP_SCHEDULE ||--|| NOTIFICATION : "triggers"
```

3.3 Referral Tracking ER Diagram

```
erDiagram
    REFERRAL {
        uuid id PK
        varchar referral_code UK
        enum status
        enum urgency
        jsonb status_history
        timestamp created_at
        timestamp completed_at
    }

    SOURCE_FACILITY {
        uuid id PK
        varchar name
        enum type
    }

    TARGET_FACILITY {
        uuid id PK
        varchar name
        enum type
    }

    SOURCE_PROVIDER {
        uuid id PK
        varchar full_name
        enum role
    }

    TARGET_PROVIDER {
        uuid id PK
        varchar full_name
        varchar specialization
    }

    PATIENT {
        uuid id PK
        varchar abha_id
        varchar full_name
    }

    ENCOUNTER {
        uuid id PK
        enum encounter_type
        text clinical_notes
    }

    NOTIFICATION_LOG {
        uuid id PK
        enum channel
        enum status
        timestamp sent_at
    }

    REFERRAL }|--|| PATIENT : "for_patient"
    REFERRAL }|--|| SOURCE_FACILITY : "referred_from"
    REFERRAL }|--|| TARGET_FACILITY : "referred_to"
    REFERRAL }|--|| SOURCE_PROVIDER : "initiated_by"
    REFERRAL }|--o| TARGET_PROVIDER : "assigned_to"
```

```
REFERRAL }|--o| ENCOUNTER : "source_encounter"  
REFERRAL }|--|{ NOTIFICATION_LOG : "generates_notifications"
```

3.4 Medicine Stock Management ER Diagram

```
erDiagram  
    MEDICINE {  
        uuid id PK  
        varchar generic_name  
        varchar brand_name  
        enum form  
        varchar strength  
        boolean is_essential  
    }  
  
    FACILITY {  
        uuid id PK  
        varchar name  
        enum type  
    }  
  
    MEDICINE_STOCK {  
        uuid id PK  
        int current_quantity  
        int reorder_level  
        date expiry_date  
        varchar batch_number  
    }  
  
    STOCK_TRANSACTION {  
        uuid id PK  
        enum transaction_type  
        int quantity  
        timestamp created_at  
    }  
  
    PRESCRIPTION_ITEM {  
        uuid id PK  
        varchar dosage  
        varchar frequency  
        int duration_days  
        boolean is_dispensed  
    }  
  
    STOCK_ALERT {  
        uuid id PK  
        enum alert_type  
        enum severity  
        boolean is_resolved  
    }  
  
    MEDICINE }|--|{ MEDICINE_STOCK : "stocked_at"  
    FACILITY }|--|{ MEDICINE_STOCK : "holds"  
    MEDICINE_STOCK }|--|{ STOCK_TRANSACTION : "logged"  
    MEDICINE_STOCK }|--o{ STOCK_ALERT : "triggers"  
    MEDICINE }|--o{ PRESCRIPTION_ITEM : "prescribed_as"  
    STOCK_TRANSACTION }|--o| PRESCRIPTION_ITEM : "dispensed_for"
```

2.6 Operational, Incentive & Supply Chain Tables

Table: asha_incentives

Automated National Health Mission (NHM) performance incentive ledger for frontline community health workers.

Column	Type	Constraints	Description
id	UUID	PK, DEFAULT gen_random_uuid()	Unique claim voucher identifier
asha_user_id	UUID	FK → users.id, NOT NULL	ASHA worker filing the claim
patient_id	UUID	FK → patients.id, NOT NULL	Associated beneficiary
encounter_id	UUID	FK → encounters.id, NULLABLE	Associated clinical encounter / referral
activity_type	ENUM	NOT NULL	JSY_DELIVERY (₹300), HRP_FOLLOWUP (₹250), VHSND_IMMUNIZATION (₹100), NCD_CBAC (₹10)
amount_inr	DECIMAL(10,2)	NOT NULL	Approved incentive amount in INR
status	ENUM	DEFAULT 'PENDING'	PENDING, VERIFIED_BY_ANM, APPROVED_BY_MO, DISBURSED
claim_date	DATE	NOT NULL	Date activity was completed in field
verified_by	UUID	FK → users.id, NULLABLE	ANM / CHO supervisor who verified field diary
disbursed_at	TIMESTAMPTZ	NULLABLE	Direct Benefit Transfer (DBT) disbursement time
created_at	TIMESTAMPTZ	DEFAULT NOW()	Record creation timestamp

Indexes: idx_asha_inc_user , idx_asha_inc_status , idx_asha_inc_date

Table: drug_inventory_batches

Batch-level pharmaceutical inventory with First-Expired, First-Out (FEFO) prioritization and e-Aushadhi / DVDMS synchronization.

Column	Type	Constraints	Description
id	UUID	PK	Unique batch entry identifier
facility_id	UUID	FK → facilities.id, NOT NULL	Facility holding the physical stock
medicine_id	UUID	FK → medicines.id, NOT NULL	Pharmaceutical master catalog reference
batch_number	VARCHAR(50)	NOT NULL	Manufacturer batch identifier
manufacture_date	DATE	NOT NULL	Manufacturing timestamp
expiry_date	DATE	NOT NULL	Expiry date for FEFO queue ranking
quantity_available	INT	NOT NULL, CHECK >= 0	Real-time units in stock
buffer_threshold	INT	DEFAULT 50	Minimum buffer level triggering inter-facility transfer
eaushadhi_batch_id	VARCHAR(100)	NULLABLE	State DVDMS sync identifier
last_synced_at	TIMESTAMPTZ	NULLABLE	Last sync timestamp with state warehouse

Indexes: `idx_drug_batch_fefo` (facility_id, medicine_id, expiry_date ASC), `idx_drug_batch_eaushadhi`

Table: `teleconsult_consent`s

Legal consent repository mandated by the NMC Telemedicine Practice Guidelines (2020) and DPDP Act 2023.

Column	Type	Constraints	Description
id	UUID	PK	Unique consent record identifier
encounter_id	UUID	FK → encounters.id, NOT NULL	Linked teleconsultation encounter
patient_id	UUID	FK → patients.id, NOT NULL	Patient granting consent
doctor_id	UUID	FK → users.id, NOT NULL	Registered Medical Practitioner on-call
consent_type	ENUM	NOT NULL	VERBAL_AUDIO_WEBRTC, SMS_OTP, BIOMETRIC, WRITTEN_PROXY
audio_recording_url	VARCHAR(500)	NULLABLE	Secure URL to encrypted verbal consent snippet
consent_timestamp	TIMESTAMPTZ	NOT NULL	Tamper-evident recording timestamp
is_valid	BOOLEAN	DEFAULT TRUE	Consent validity flag
created_at	TIMESTAMPTZ	DEFAULT NOW()	

Indexes: `idx_tc_consent_encounter` , `idx_tc_consent_patient`

4. Technical Data Flow — Main ER Diagram

This diagram shows the complete data flow across the platform from patient entry to state-level analytics.

flowchart TD

```
subgraph VILLAGE["VILLAGE LEVEL"]
    P[Patient] --> ASHA[ASHA Worker App]
    ASHA --> REG[Patient Registration<br/>ABHA ID Creation]
    ASHA --> HV[Home Visit Data<br/>CBAC + Vitals]
    ASHA --> ASHA_INC[NHM Incentive Ledger<br/>Auto-Claim Voucher]
end
```

```
subgraph SC["SUB-CENTRE / AAM"]
    HV --> TRIAGE[4-Tier CDSS Triage Engine]
    TRIAGE --> |Emergency Red| EMR[108 Emergency Ambulance]
    TRIAGE --> |Urgent Amber| REF1[Priority Referral to CHC]
    TRIAGE --> |Semi-Urgent Yellow| TC1[PHC Teleconsult Request]
    TRIAGE --> |Routine Green| SC_ADV[Local AAM Dispense & VHSND]

    REG --> LPR[Longitudinal Patient Record<br/>CBOR Delta Sync]
    HV --> LPR
end
```

```
subgraph PHC["PRIMARY HEALTH CENTRE"]
    REF1 --> APT[Appointment + Token]
    APT --> QUEUE[Queue Management]
    QUEUE --> ENC[Doctor Encounter]
    TC1 --> TC_SESSION[NMC Teleconsult Session<br/>WebRTC Audio/Video]
    TC_SESSION --> CONSENT[Teleconsult Consent Log]
    TC_SESSION --> ENC

    ENC --> RX[e-Prescription Format Appx 2]
    ENC --> DX_ORD[Diagnostic Order]
    ENC --> |Tertiary Need| REF2[Referral to CHC/DH]
    ENC --> FU[Follow-Up Schedule]

    RX --> DISP[Dispensary FEFO Allocation]
    DISP --> STOCK[e-Aushadhi / Batch Stock Update]

    DX_ORD --> |If available locally| LAB[Lab Processing]
    DX_ORD --> |If not available| DX_COORD[Diagnostic Coordination<br/>Route to Nearest Facility]

    LAB --> DX_RES[Diagnostic Results]
    DX_COORD --> DX_RES

    DX_RES --> LPR
    RX --> LPR
    ENC --> LPR
end
```

```
subgraph CHC["COMMUNITY HEALTH CENTRE / DISTRICT HOSPITAL"]
    REF2 --> REF_ACCEPT[Referral Acceptance]
    REF2 --> MJPJAY[MJPJAY Insurance Package<br/>Pre-Authorization]
    REF_ACCEPT --> SPEC_APT[Specialist Appointment]
    SPEC_APT --> SPEC_ENC[Specialist Encounter]
    SPEC_ENC --> TREAT[Treatment Plan]
    TREAT --> LPR

    EMR --> EMRG_ENC[Emergency Trauma/Envenomation]
    EMRG_ENC --> LPR
end
```

```
subgraph NOTIFY["NOTIFICATION ENGINE"]
    FU --> N_SMS[SMS/WhatsApp Reminders]
    REF2 --> N_REF[Referral Status Updates]
    STOCK --> N_STOCK[FEFO Stock Alert Notifications]
    APT --> N_APT[Appointment Reminders]
end
```

```
end

subgraph DASHBOARD["DASHBOARDS & ANALYTICS"]
  LPR --> F_DASH[Facility Dashboard]
  F_DASH --> D_DASH[District DHO Dashboard]
  D_DASH --> S_DASH[State MSInS/Health Dashboard]

  F_DASH --> KPI[7 Core M&E KPIs:<br/>RCR, MTTC, HRP Tracking,<br/>Stockout, ASHA Velocity, Mesh Syn
end

subgraph ABDM["ABDM & STATE INTEGRATION"]
  LPR --> FHIR[FHIR R4 Resource<br/>Generation]
  FHIR --> ABDM_GW[ABDM Gateway M1/M2/M3<br/>ABHA, HFR, HPR, Consent]
  STOCK --> EAUSHADHI[e-Aushadhi / DVDMS<br/>State Drug Warehouse]
end

style VILLAGE fill:#e8f5e9,stroke:#2e7d32
style SC fill:#e3f2fd,stroke:#1565c0
style PHC fill:#fff3e0,stroke:#e65100
style CHC fill:#fce4ec,stroke:#c62828
style NOTIFY fill:#f3e5f5,stroke:#6a1b9a
style DASHBOARD fill:#e0f2f1,stroke:#00695c
style ABDM fill:#ede7f6,stroke:#4527a0
```

5. Summary Statistics

Metric	Count
Total Tables	25 (Expanded with ASHA Incentives, FEFO Batches, and Legal Consents)
Core Entity Tables	5 (users, patients, facilities, encounters, referrals)
Clinical Tables	7 (vitals, triage, prescriptions, prescription_items, diagnostic_orders, diagnostic_results, teleconsult_consents)
Support & Supply Chain Tables	9 (medicines, stock, stock_transactions, drug_inventory_batches, equipment, appointments, queue, follow_ups, asha_incentives)
System & Audit Tables	4 (notifications, audit_log, sync_queue, user_sessions, facility_daily_stats)
Total Relationships	52+ foreign key relationships with strict referential integrity
FHIR R4 Aligned Entities	8 (Patient, Encounter, Observation, MedicationRequest, ServiceRequest, DiagnosticReport, Appointment, Organization)

Database schema designed for ArogyaSetu Bridge — SIH 2026, PS 26133
All diagrams use Mermaid notation for portability and version control.